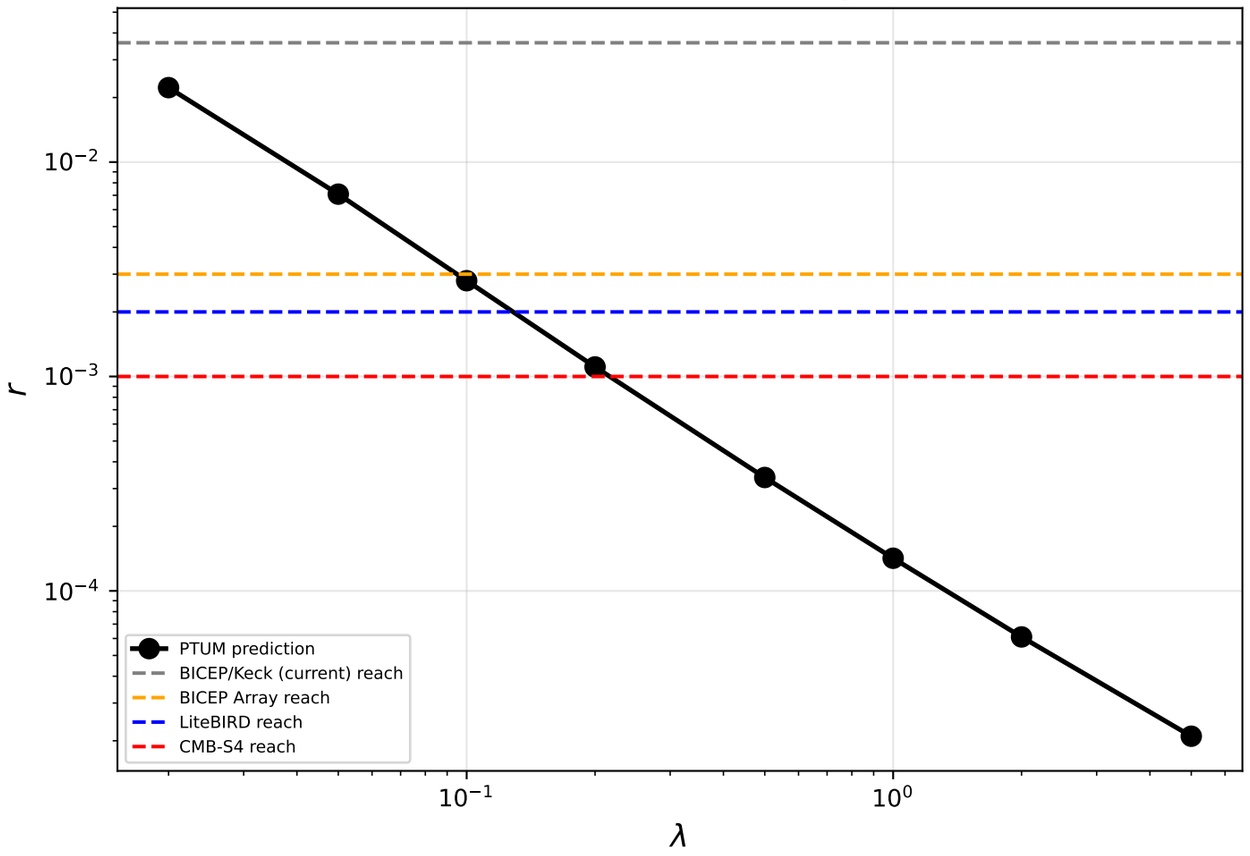
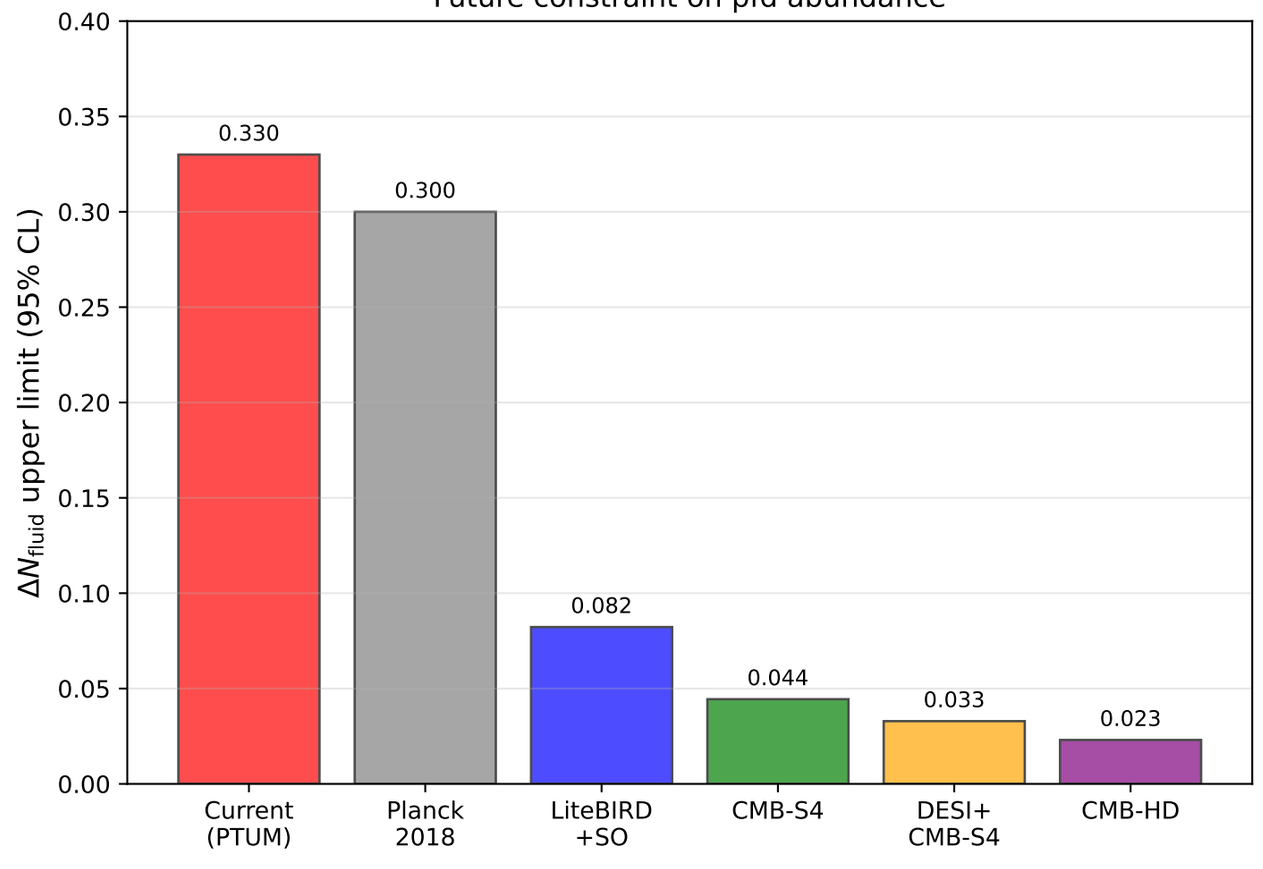
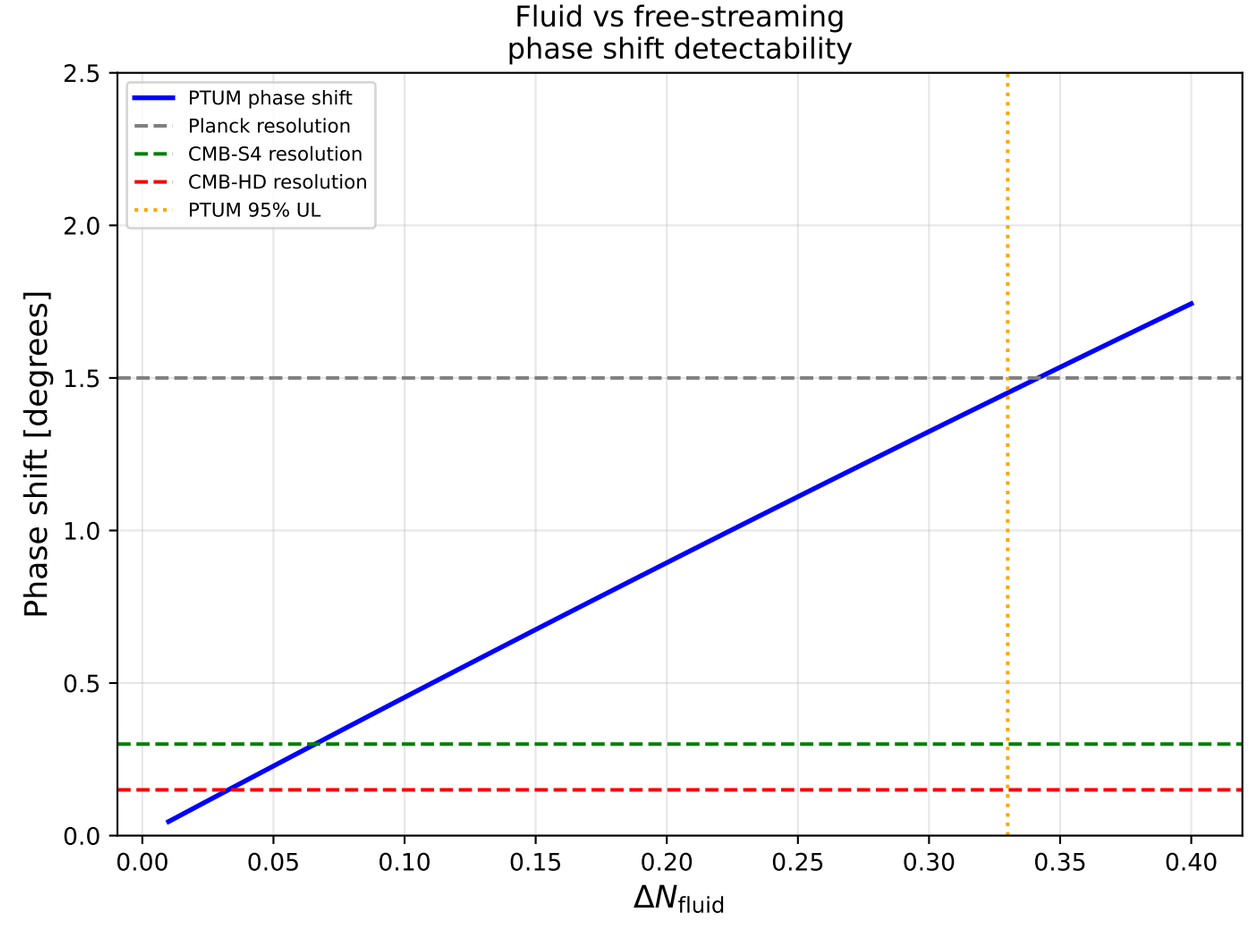
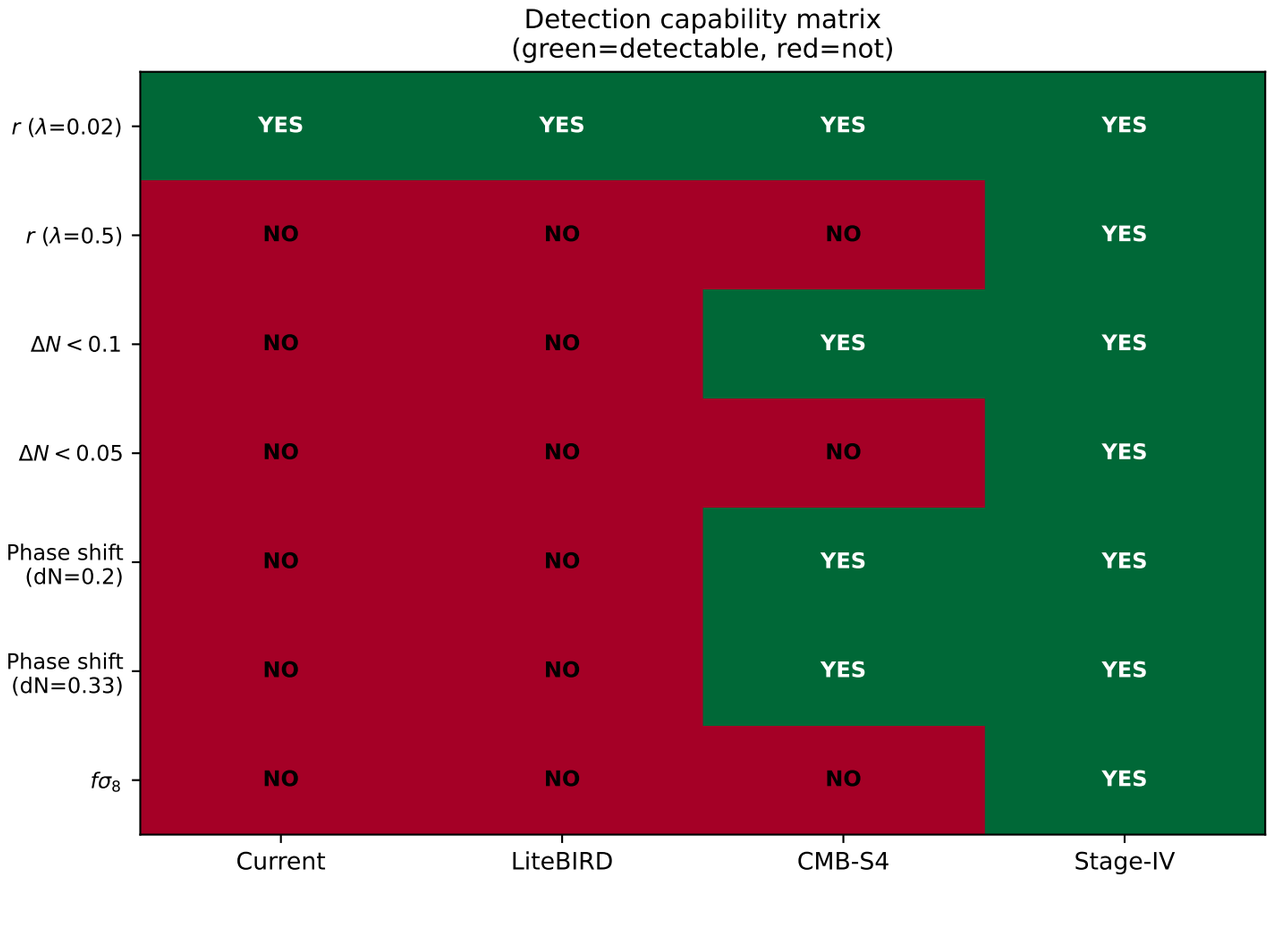

Tensor ratio detectability
10^{-2}
10^{-3}
10^{-4}
r
10^{-1}
10^{0}
λ
PTUM prediction
BICEP/Keck (current) reach
BICEP Array reach
LiteBIRD reach
CMB-S4 reach


Future constraint on pfd abundance
$\Delta N_{\rm fluid}$ upper limit (95% CL)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0.330
0.300
0.082
0.044
0.033
0.023
Current (PTUM)
Planck 2018
LiteBIRD +SO
CMB-S4
DESI+ CMB-S4
CMB-HD


Fluid vs free-streaming
phase shift detectability
Phase shift [degrees]
0.0
0.5
1.0
1.5
2.0
2.5
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
$\Delta N_{\rm fluid}$
PTUM phase shift
Planck resolution
CMB-S4 resolution
CMB-HD resolution
PTUM 95% UL


Detection capability matrix
(green=detectable, red=not)
r (λ=0.02)
r (λ=0.5)
$\Delta N < 0.1$
$\Delta N < 0.05$
Phase shift (dN=0.2)
Phase shift (dN=0.33)
$f\sigma_8$
YES
NO
Current
LiteBIRD
CMB-S4
Stage-IV